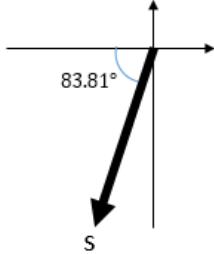
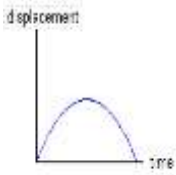
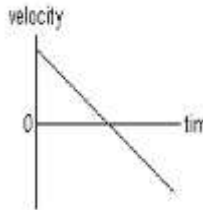
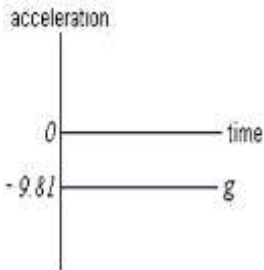
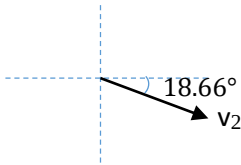
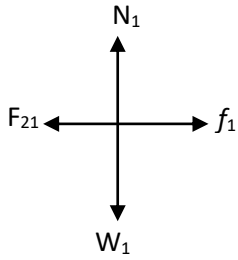
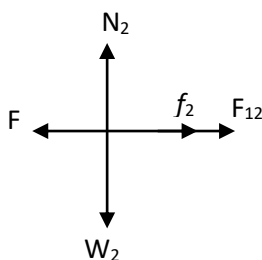


ANSWER SCHEME TRIAL 1

a)	$u = \frac{60 \text{ km}}{1 \text{ hr}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ hr}}{3600 \text{ s}}$ $= 16.67 \text{ ms}^{-1}$ $v = \frac{120 \text{ km}}{1 \text{ hr}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ hr}}{3600 \text{ s}}$ $= 33.33 \text{ ms}^{-1}$ $a = \frac{v - u}{t} = \frac{33.33 - 16.67}{5} = 3.33 \text{ ms}^{-2}$	J1 J1 GJU1															
b)	<table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <thead> <tr> <th>Vector</th><th>x-component (m)</th><th>y-component (m)</th></tr> </thead> <tbody> <tr> <td>A</td><td>$A_x = -A$ $= -15$</td><td>$A_y = 0$</td></tr> <tr> <td>B</td><td>$B_x = -B \cos \theta_1$ $= -20 \cos 45$ $= -14.14$</td><td>$B_y = -B \sin \theta_1$ $= -20 \sin 45$ $= -14.14$</td></tr> <tr> <td>C</td><td>$C_x = C \sin \theta_2$ $= 30 \sin 60$ $= 25.98$</td><td>$C_y = -C \cos \theta_2$ $= -30 \cos 60$ $= -15$</td></tr> <tr> <td>Resultant</td><td>-3.16</td><td>-29.14</td></tr> </tbody> </table> $s = \sqrt{s_x^2 + s_y^2}$ $= \sqrt{(-3.16)^2 + (-29.14)^2}$ $= 29.31 \text{ m}$ $\theta = \tan^{-1} \left \frac{s_y}{s_x} \right $ $= \tan^{-1} \left \frac{-29.14}{-3.16} \right $ $= 83.81^\circ \text{ below negative } x - \text{axis}$ @ provide diagram 	Vector	x-component (m)	y-component (m)	A	$A_x = -A$ $= -15$	$A_y = 0$	B	$B_x = -B \cos \theta_1$ $= -20 \cos 45$ $= -14.14$	$B_y = -B \sin \theta_1$ $= -20 \sin 45$ $= -14.14$	C	$C_x = C \sin \theta_2$ $= 30 \sin 60$ $= 25.98$	$C_y = -C \cos \theta_2$ $= -30 \cos 60$ $= -15$	Resultant	-3.16	-29.14	$s_x = J1,$ $s_y = J1$ GJU1 Angle = J1 Direction = J1
Vector	x-component (m)	y-component (m)															
A	$A_x = -A$ $= -15$	$A_y = 0$															
B	$B_x = -B \cos \theta_1$ $= -20 \cos 45$ $= -14.14$	$B_y = -B \sin \theta_1$ $= -20 \sin 45$ $= -14.14$															
C	$C_x = C \sin \theta_2$ $= 30 \sin 60$ $= 25.98$	$C_y = -C \cos \theta_2$ $= -30 \cos 60$ $= -15$															
Resultant	-3.16	-29.14															

2.(a)	i)  displacement time ii)  velocity time iii)  acceleration time -9.81 g	D2 x 3 = 6 (The pattern and label of axis must be correct)
2.(b)(i)	At max height, $v = 0 \text{ m s}^{-1}$ $v^2 = u^2 - 2gs$ From: $(0)^2 = (20)^2 - 2(9.81)s$ $s = \frac{400}{19.62}$ $= 20.39 \text{ m}$	K1 G1 JU1
2.(b)(ii)	At max height, $v = 0 \text{ m s}^{-1}$ $v = u - gt$ $0 = 20 - 9.81t$ $t = 2.04s$	G1 JU1
2.(b)(iii)	Given : $s = -25 \text{ m}$ (below initial point) From : $s = ut - \frac{1}{2}gt^2$ $-25 = 20t - \frac{1}{2}9.81t^2$ $t = 5.1s$	G1 JU1

3(a)	Given; $m_1 = 8.0 \text{ kg}$, $u_1 = 3.35 \text{ m s}^{-1}$, $m_2 = 5.0 \text{ kg}$, $u_2 = 0 \text{ m s}^{-1}$ $v_1 = -5.45 \text{ m s}^{-1}$, 30° above horizontal. $v_2 = ?$ i) $\sum P_{i_x} = \sum P_{f_x}$ $m_1 u_{1x} + m_2 u_{2x} = m_1 v_{1x} + m_1 v_{2x}$ $8.0(3.35) + 5(0) = (8.0)(-5.45 \cos 30^\circ) + (5.0)v_{2x}$ $v_{2x} = 12.91 \text{ m s}^{-1}$ $\sum P_{i_y} = \sum P_{f_y}$ $m_1 u_{1y} + m_2 u_{2y} = m_1 v_{1y} + m_1 v_{2y}$ $0 = (8.0)(5.45 \sin 30^\circ) + (5.0)v_{2y}$ $v_{2y} = -4.36 \text{ m s}^{-1}$ $ v_2 = \sqrt{v_{2x}^2 + v_{2y}^2}$ $ v_2 = \sqrt{(12.91)^2 + (-4.36)^2}$ $\therefore v_2 = 13.63 \text{ m s}^{-1}$ $\tan \theta = \frac{v_{2y}}{v_{2x}}$ $\therefore \theta = \tan^{-1} \frac{(-4.36)}{(12.91)} = -18.66^\circ$ The direction is 18.66° below positive x-axis Or 	} K1 GJ1 GJ1 K1 GJU1 K1 JU1

(b)	m= 100 g = 0.1 kg ; u =12.0 m s⁻¹ ; v = - 8.0 m s⁻¹ Impulse of the ball on the wall, J =? $J = \Delta p$ $J = m(v - u)$ $J = 0.1(-8.0 - 12.0)$ $\therefore J = -2.0 \text{ kg m s}^{-1}$ or $J = -2.0 \text{ N s}$	G1 JU1
4 (i)	For m₁  Label all forces correctly – 1 m Correct direction of arrow – 1m For m₂  Label all forces correctly – 1 m Correct direction of arrow – 1m (ii) $f_1 = \mu N_1$ $f_1 = (0.4)(0.3)(9.81) = 1.177 \text{ N}$ $f_2 = \mu N_2$ $f_2 = (0.4)(0.5)(9.81) = 1.962 \text{ N}$ (iii) $\sum F_x = ma$ $F + (-f_1) + (-f_2) = ma$ $F = ma + f_1 + f_2$	D2 D2 GJU1 GJU1 K1 G1

(iv)		$F = (0.8)(2) + 1.177 + 1.962$ $F = 4.739 \text{ N}$ $\sum F_x = m_2 a$ $F + (-f_2) + (-F_{12}) = m_2 a$ $F_{12} = 4.739 - 1.962 - (0.5)(2)$ $F_{12} = 1.78 \text{ N}$	JU1 K1 G1 JU1
5. a)		$T = \frac{2\pi}{\omega}$ $\omega = \frac{2\pi}{T} = \frac{2\pi}{360} = 0.0175 \text{ rads}^{-1}$ $r = 2.6 \text{ km} = 2.6 \times 10^3 \text{ m}$ $a_c = r\omega^2 = (2.6 \times 10^3)(0.0175)^2$ $= 0.796 \text{ ms}^{-2}$	K1 G1 JU1
b)		Horizontal component: $T \sin \theta = \frac{mv^2}{r}$ (1) Vertical component: $T \cos \theta - mg = 0$ (2) Solving (1) / (2) $\tan \theta = \frac{v^2}{rg}$ $v = \sqrt{rg \tan \theta}$ $r = L \sin \theta = 0.32 \sin 30 = 0.16 \text{ m}$ Hence, $v = \sqrt{0.16 (9.81) \tan 30^\circ}$ $= 0.95 \text{ ms}^{-1}$	K1 K1 K1 K1 G1 JU1
6. a)	i)	The angular speed, $\omega = \frac{1200 \text{ rev}}{1 \text{ min}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} \times \frac{1 \text{ min}}{60 \text{ s}} = 125.66 \text{ rad s}^{-1}$	K 1 GJU 1
	ii)	the tangential speed at a point 3.00 cm from its centre, $v = r\omega$ $v = (0.03)(125.66)$ $v = 3.77 \text{ ms}^{-1}$	G 1 JU 1

iii)	the radial acceleration of a point on the rim, $a_c = r\omega^2$ $a_c = (0.08)(125.66)^2$ $a_c = 1263.23 \text{ ms}^{-2} = 1.26 \times 10^3 \text{ ms}^{-2}$	G 1 JU 1								
b)	The resultant (magnitude and direction) torque of all forces about point O, <table border="1"> <thead> <tr> <th>Force (N)</th> <th>Torque (N m), $\tau = Fd = Fr\sin\theta$</th> </tr> </thead> <tbody> <tr> <td>$\vec{F}_1 = 30$</td> <td>$\tau_1 = -(30)(3) \sin 90^\circ = -90$</td> </tr> <tr> <td>$\vec{F}_2 = 10$</td> <td>$\tau_2 = (10)(5) \sin 0^\circ = 0$</td> </tr> <tr> <td>$\vec{F}_3 = -20$</td> <td>$\tau_3 = +(20)(5) \sin 90^\circ = +100$</td> </tr> </tbody> </table> The resultant torque: $\sum \tau_O = \tau_1 + \tau_2 + \tau_3$ $\sum \tau_O = -90 + 0 + 100 = +10 \text{ N m}$ (anti-clockwise)	Force (N)	Torque (N m), $\tau = Fd = Fr\sin\theta$	$\vec{F}_1 = 30$	$\tau_1 = -(30)(3) \sin 90^\circ = -90$	$\vec{F}_2 = 10$	$\tau_2 = (10)(5) \sin 0^\circ = 0$	$\vec{F}_3 = -20$	$\tau_3 = +(20)(5) \sin 90^\circ = +100$	G 1 G 1 G 1 K 1 JU 1 J 1
Force (N)	Torque (N m), $\tau = Fd = Fr\sin\theta$									
$\vec{F}_1 = 30$	$\tau_1 = -(30)(3) \sin 90^\circ = -90$									
$\vec{F}_2 = 10$	$\tau_2 = (10)(5) \sin 0^\circ = 0$									
$\vec{F}_3 = -20$	$\tau_3 = +(20)(5) \sin 90^\circ = +100$									
7.(a)(i)	Temperature gradient = $\frac{dT}{dx} = \frac{T_f - T_i}{x}$ $= \frac{0 - 60}{4}$ $= -15 \text{ }^\circ\text{C m}^{-1}$	G1 JU1								
7.(a)(ii)	$\frac{dQ}{dt} = -kA \frac{dT}{dx}$ $\frac{dQ}{dt} = -(209)(16)(-15)$ $= 50160 \text{ W / J s}^{-1}$	G1 JU1								
7.(a)(iii)	$dQ = -kA \frac{dT}{dx} (dt)$ $= 50160 (2)$ $= 1.00 \times 10^5 \text{ J}$ @	G1 JU1								

	$dQ = -(209)(16)(-15)(2)$ $= 1.00 \times 10^5 \text{ J}$ Hence, the amount of heat that flows through the cube is $1.00 \times 10^5 \text{ J}$.	
7.(b)(i)	Temperature, $T = 20^\circ\text{C} = 40 + 273.15$ $= 313.15 \text{ K}$	G1 JU1
7.(b)(ii)	$R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1},$ $v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$ $v_{\text{rms}} = \sqrt{\frac{3(8.31)(313.15)}{0.032}} = 493.93 \text{ m s}^{-1}$ V_{rms} for Nitrogen gas is 493.93 m s^{-1}	K1 GJU1
7.(c)(i)	$\Delta U_{AB} = -300 \text{ J}$ $W_{AB} = -135 \text{ J}$ $Q_{AB} = \Delta U_{AB} + W_{AB}$ $Q_{AB} = -300 - 135$ $Q_{AB} = -435 \text{ J}$ $\therefore$ Heat is released.	G1 JU1 J1
7.(c)(ii)	Work done in BC, $W_{BC} = 0 \text{ J}$ (isochoric, V: constant) $Q_{BC} = 160 \text{ J}$ $Q_{BC} = \Delta U_{BC} + W_{BC}$ $160 = \Delta U_{BC} + 0$ $\Delta U_{BC} = 160 \text{ J}$	J1 GJU1
7.(c)(iii)	$\Delta U_{CA} = 0 \text{ J}$ (isothermal, T : constant) $W_{CA} = 450 \text{ J}$ $Q_{CA} = \Delta U_{CA} + W_{CA}$ $Q_{CA} = 0 + 450$ $Q_{CA} = 450 \text{ J}$	J1 GJU1